

Submit your Jupyter notebook with the answers to gradescope by 3:40pm.

1. Let `s = lambda n: float(sum([exp(-(k/n)**2)*sin(2*k/n) for k in range(n)]))/n`

be a function to approximate $\int_0^1 \exp(-x^2) \sin(2x) dx$.

- (a) Define a Python function `p` which does the same as `s`.

For correctness, compare the output of `p(1000)` and `s(1000)`.

- (b) Apply Cython to improve the efficiency of `p`.

Time the execution of the Cython function for $n = 10000$, compare with timings of `p`.

2. Let $p = 6x^4y^3 - x^2y^6 + x^2y^3 + x^2 - 2y^7 - y^3$.

Without retyping p , convert p into $-x^2y^6 + 6x^4y^3 - 2y^7 + x^2y^3 - y^3 + x^2$.